

Generalized P, Q and G Conditions

Brian Weiner

Pennsylvania State University

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Abstract

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I. INTRODUCTION

The N -Representability Problem is to find the *necessary and sufficient conditions* that a positive trace class operator, that acts on two electron space, must satisfy in order to be the contraction of a N -electron state density operator and thus be a Second Order Reduced Density Operator (SORDO). It was first precisely formulated in quantum chemistry [1] by Coleman as a result of Coulsons *challenge* [2], which was inspired by work initiated by Husimi [3]. The significance of the problem derives from the realization that the properties of molecular systems, at a fixed geometry, are fully described by Hamiltonians that describe the kinetic energy of the individual electrons, interactions of the electrons with external and internal electric and magnetic fields and *only* the simultaneous pair wise interactions of the

electrons with each other. As simultaneous three, four and higher body interactions are not significant the N -electron state density operator contains redundant information and the energy of molecular systems can be fully expressed in terms of the SORDO of the system, which depends on fewer variables. Hence the determination of the ground state energy can be formulated as a minimization of a linear functional over operators that act in two electron space that are *constrained* to be SORDO's, instead of a higher dimensional minimization over N -electron state density operators.

An intrinsic solution (i.e. only in terms of properties of operators acting in two electron space) has not been found despite a large amount of effort [4]. This work has, however, resulted in the characterization of numerous necessary conditions (see for instance [5]), the innovative contracted Schrödinger equation method [6], the relation to complexity theory [7] and various variants of the problem i.e. pure/ensemble, particle conserving/ particle nonconserving and finite/infinite dimensional.

In a paper [8] the N -Representability Problem and the less restrictive Representability Problem was shown to be one of finding conditions for the existence of extensions of a linear functional from a subspace to a whole vector space, placing it firmly in the area of mathematical analysis and allowing one to utilize theorems proved in this topic over the last hundred years. The classical Hahn-Banach theorem [9] on extensions, $\tilde{f}$, of unrestricted linear functionals, f , defined on subspaces, W , to the whole vector space V was specialized to positive linear functionals defined on ordered vector spaces by Krein [10] and to self-adjoint subspaces of c^* -algebras that *might or might not* contain the identity by Segal [11]. In order for the extension to be a state [9] i.e. a positive normalized linear functional on the whole algebra its value on the identity must be one i.e.

$$\tilde{f}(I) = 1 \tag{1}$$

so if $I \in W$ then

$$\tilde{f}(I) = f(I) = 1. \tag{2}$$

Segal's theorem examines the implications of this normalization and shows that a positive normalized extension $\tilde{f}$ exists if (a)

$$f(X) \geq 0 \forall X \in W_+, \tag{3}$$

where

$$W_+ = \{X \in W \mid X \geq 0\}, \quad (4)$$

(b)

$$\sup_{X \in W_<} f(X) \leq 1, \quad (5)$$

where the positive part of the subspace W that contains operators that are less than or equal to the identity operator is defined as

$$W_< = \{X \in W \mid I \geq X \geq 0\} \quad (6)$$

and (c)

$$\inf_{X \in W_>} f(X) \geq 1, \quad (7)$$

where the positive part of the subspace W that contains operators that are greater than or equal to the identity operator is defined as

$$W_> = \{X \in W \mid X \geq I\}. \quad (8)$$

It can easily be seen that if $X \geq 0$ then (Appendix A)

$$\|X\| \leq 1 \Leftrightarrow I \geq X \geq 0 \quad (9)$$

and thus

$$= \{X \in W \mid X \in W_+; \|X\| \leq 1\}, \quad (10)$$

so $W_<$ is in the positive part of the unit ball of the algebra and $W_>$ belongs to the annulus

$$W_> = \{X \in W \mid X \geq I; \|X\| \geq 1\}. \quad (11)$$

The unit ball is a convex set and the Krein-Millmann theorem [9] shows that in a w^* -algebra [9] it is equal to the closure of its extreme points [12], moreover the positive part of the unit ball in a w^* -algebra also forms a convex set which is also the closure of *its* extreme points. In an important theorem Kadison [13] has shown that the extreme points of the positive part of the unit ball in a w^* -algebra are projectors and in particular belong to the unit sphere

$$\|X\| = 1. \quad (12)$$

In this paper the subspaces we need to consider do not contain any operators that belong to $W_>$ thus we only need to take into account the conditions eqs. (3) and (5). The set W_+

is also a convex set and is generated by its extreme points, $\text{ext } \{W_+\}$, which is the set of projectors in W . A functional is positive on W if and only if it is positive on $\text{ext } \{W_+\}$, while the supremum of the values of linear functionals defined on convex sets only occur at extreme points of the set, thus one has only to consider the values of the functionals on the projectors $\text{ext } \{W_+\}$ in order to ascertain whether they satisfy both conditions eq. (5) and eq. (3). The *necessary and sufficient* conditions for the existence of a state that extends a linear functional defined on a subspace W can thus be expressed in terms of values of the functional on the set of projectors $\text{ext } \{W_+\}$ as

$$1 \geq f(X) \geq 0, \quad \forall X \in \text{ext } \{W_+\}. \quad (13)$$

In the context of the representability problem for electrons the relevant w^* -algebra is the Fock algebra and the subspace is the one formed by one and two electron operators, while the N -Representability Problem can be expressed by the addition of further constraints. The lower limits in eq. (13) correspond to the well known [14] *necessary* P, Q and G conditions, however the upper limits in eq. (13) correspond to the Krein, Segal and Kadison theorems, that in conjunction with the lower limits, make the conditions *necessary and sufficient* for ensemble representability and we describe them as *generalized* P, Q and G conditions.

In the following we discuss and examine the preceding ideas first reviewing, in section II, some mathematical concepts and notation that are utilized in the subsequent sections then in section III the expansion and contraction maps, which are adjoints of each other, are considered in the context of Fock space [12, 15], which leads to the definition of Reduced Density Operators (RDO's). In section IV we utilize the fact that the Fock algebra is a concrete realization of a w^* -algebra and show that RDO's correspond to Reduced States (RS's) that are positive functionals defined on self adjoint subspaces of a w^* -algebra that can be extended to states of the algebra. This shows that the representability problem can be formulated as finding the conditions that positive linear functionals defined on subspaces must satisfy in order to be extended to states. Kadisons theorem is then used to show that *necessary and sufficient* extension conditions can be expressed in terms of values of the functionals on projectors contained in the subspace. In section V the form of *all* projectors that can be expressed as linear combinations of one and two particle operators is found. In section VI we display that the conditions found from Kadisons theorem lead to generalizations of the known *necessary* P, Q and G conditions for representability and can

be expressed in terms of matrix elements of first and second order reduced density matrices. In section VII we discuss the condition that must be satisfied to make the functionals and reduced density matrices ensemble N -representable, which leads to the list of *necessary and sufficient* ensemble N -representability conditions given in section VIII and we conclude with a discussion in section IX.

II. MATHEMATICAL ESSENTIALS

In the following the dimension of all spaces are considered to be finite and can be thought of as approximations to the infinite dimensional case, which must be used to obtain the exact solutions of quantum mechanical equations. However notation that is appropriate for the infinite dimensional case will be used even though it becomes redundant in the finite dimensional case.

A. Fermi Fock Space

We consider the Fermi Fock space, $\mathcal{H}_F$, based on the one electron Hilbert space $\mathcal{H}^1$, defined by the direct sum

$$\mathcal{H}_F = \bigoplus_{0 \leq N \leq r} \mathcal{H}^N \quad (14)$$

of the N -electron Hilbert spaces

$$\mathcal{H}^N = \bigwedge^N \mathcal{H}^1, \quad (15)$$

where $\wedge$ denotes antisymmetric tensor product and $\mathcal{H}^0 = \{ \lambda |\phi\rangle | \lambda \in \mathbb{C} \}$, and $|\phi\rangle$ is the vacuum state. The inner product in the Hilbert space $\mathcal{H}_F$ is defined as

$$\langle \Psi | \Phi \rangle_{\mathcal{F}} = \sum_{0 \leq N \leq r} \langle \Psi_N | \Phi_N \rangle_{\mathcal{H}^N} = \sum_{0 \leq N \leq r} \langle \Psi | P_{\mathcal{H}^N} \Phi \rangle_{\mathcal{H}^N}, \quad (16)$$

where $P_{\mathcal{H}^N}$ is the orthogonal projector onto $\mathcal{H}^N$.

B. Fermi Fock Algebra

The fermion Fock algebra is the space of bounded operators (Appendix B) acting in $\mathcal{H}_F$ i.e.

$$\mathcal{F} = \mathcal{B}(\mathcal{H}_F). \quad (17)$$

This space can be generated by polynomials

$$X = \sum_{0 \leq P, Q \leq r} \sum_{\substack{1 \leq j_1 < \dots < j_P \leq r \\ 1 \leq k_1 < \dots < k_Q \leq r}} X_{j_1 \dots j_P k_1 \dots k_Q} a_{j_1}^\dagger \dots a_{j_P}^\dagger a_{k_Q} \dots a_{k_1} \quad (18)$$

of second quantized operators, $\left\{ a_j^\dagger \mid 1 \leq j \leq r \right\}$, which are defined by their action on $|\phi\rangle$ by

$$a_j^\dagger |\phi\rangle = |\varphi_j\rangle, 1 \leq j \leq r, \quad (19)$$

where $\{|\varphi_j\rangle \mid 1 \leq j \leq r\}$ is a complete orthonormal basis of one-particle space $\mathcal{H}^1$ and satisfy the anti-commutation relationships given by

$$\left[a_j^\dagger, a_k \right]_+ = a_j^\dagger a_k + a_k a_j^\dagger = \delta_{jk}, 1 \leq j, k \leq r. \quad (20)$$

The fermion Fock algebra can be decomposed as a vector space direct sum

$$\mathcal{F} = \mathcal{F}_e \oplus \mathcal{F}_0 \quad (21)$$

of a particle conserving sub algebra $\mathcal{F}_e$ and a particle non conserving subspace of operators $\mathcal{F}_0$. The sub algebra $\mathcal{F}_e$ can be further decomposed as a direct sum of sub spaces

$$\mathcal{F}_e = \bigoplus_{0 \leq N \leq r} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_N, \quad (22)$$

where

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}})_N = \left\{ \sum_{\substack{1 \leq j_1 < \dots < j_N \leq r \\ 1 \leq k_1 < \dots < k_N \leq r}} X_{j_1 \dots j_N k_1 \dots k_N} a_{j_1}^\dagger \dots a_{j_N}^\dagger a_{k_N} \dots a_{k_1} \right\} \quad (23)$$

or as a direct sum of sub algebras

$$\mathcal{F}_e = \bigoplus_{0 \leq N \leq r} \mathcal{B}(\mathcal{H}^N), \quad (24)$$

where $\mathcal{B}(\mathcal{H}^N)$ is the space of bounded operators acting in $\mathcal{H}^N$. The sub space $\mathcal{F}_0$ can be decomposed as the vector space direct sum

$$\mathcal{F}_0 = \bigoplus_{0 \leq N \neq M \leq r} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{N,M}, \quad (25)$$

where

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{N,M} = \left\{ \sum_{\substack{1 \leq j_1 < \dots < j_N \leq r \\ 1 \leq k_1 < \dots < k_M \leq r}} X_{j_1 \dots j_N k_1 \dots k_M} a_{j_1}^\dagger \dots a_{j_N}^\dagger a_{k_M} \dots a_{k_1} \right\} \quad (26)$$

or as the vector space direct sum

$$\mathcal{F}_o = \bigoplus_{0 \leq N \neq M \leq r} \mathcal{B}(\mathcal{H}^N, \mathcal{H}^M), \quad (27)$$

where $\mathcal{B}(\mathcal{H}^N, \mathcal{H}^M)$ is the vector space of bounded linear maps from the Hilbert space $\mathcal{H}^N$ to the Hilbert space $\mathcal{H}^M$. The decompositions eqs. (22) and (25) express the fermion Fock algebra in *second quantized* form as polynomials of second quantized operators, while the decompositions eqs. (24) and (27) express it in *first quantized* form as expansions of the ket-bras

$$\{|\varphi_{j_1} \cdots \varphi_{j_N}\rangle \langle \varphi_{k_1} \cdots \varphi_{k_M}| \mid 1 \leq j_1 < \cdots < j_N \leq r, 1 \leq k_1 < \cdots < k_M \leq r; 0 \leq N, M \leq r\} \quad (28)$$

.

C. Operator Inner Product

The Fock space trace operation is defined as

$$\text{Tr}\{X\} = \sum_{0 \leq N \leq r} \sum_{1 \leq j_1 < \cdots < j_N \leq r} \langle \Psi_{Nj_1 \cdots j_N} | X \Psi_{Nj_1 \cdots j_N} \rangle, \quad (29)$$

where $\{\Psi_{Nj_1 \cdots j_N} \mid 1 \leq j_1 < \cdots < j_N \leq r; 0 \leq N \leq r\}$ are complete orthonormal bases for $\mathcal{H}^N, 0 \leq N \leq r$. One can use this trace operation to define the space of trace class operators $\mathcal{B}_1(\mathcal{H}_{\mathcal{F}})$ that have finite trace for all bases and the space of Hilbert Schmidt operators $\mathcal{B}_2(\mathcal{H}_{\mathcal{F}})$ that form a Hilbert space with inner product

$$(X|Y) = \text{Tr}\{X^\dagger Y\}. \quad (30)$$

In finite dimensions, i.e. $r < \infty$, there is no distinction between these spaces of operators i.e. $\mathcal{B}(\mathcal{H}_{\mathcal{F}}) = \mathcal{B}_1(\mathcal{H}_{\mathcal{F}}) = \mathcal{B}_2(\mathcal{H}_{\mathcal{F}})$.

III. EXPANSION AND CONTRACTION MAPS

The convex set, $S_{\mathcal{F}}$, of density operators in $\mathcal{F}$ is given by

$$S_{\mathcal{F}} = \{D \mid D \in \mathcal{B}_1(\mathcal{H}_{\mathcal{F}}); D \geq 0; \text{Tr } D = 1\}, \quad (31)$$

whose elements could describe any type of physical state i.e. electron conserving or non conserving, ensemble or pure etc.

A. Contraction Map

Employing the work of [15] one can define a *non singular* generalized contraction map

$$L : \mathcal{B}_1(\mathcal{H}_{\mathcal{F}}) \rightarrow \mathcal{B}_1(\mathcal{H}_{\mathcal{F}}) \quad (32)$$

by

$$L(Y) = \sum_{0 \leq N, M \leq r} \sum_{\substack{1 \leq j_1 < \dots < j_M \leq r \\ 1 \leq k_1 < \dots < k_N \leq r}} \left(a_{k_1}^\dagger \dots a_{k_N}^\dagger a_{j_M} \dots a_{j_1} \middle| Y \right) |\varphi_{j_1} \dots \varphi_{j_M}\rangle \langle \varphi_{k_1} \dots \varphi_{k_N}|. \quad (33)$$

B. Expansion Map

The expansion map

$$\Gamma : \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \quad (34)$$

is adjoint to the contraction map, i.e. $\Gamma = L^\dagger$, with respect to inner product $(\cdot | \cdot)$ and is defined by

$$(\Gamma(X) | Y) = (X | L(Y)) \quad \forall Y \in \mathcal{B}_1(\mathcal{H}_{\mathcal{F}}). \quad (35)$$

One can show (Appendix SecQuant) that Γ is *actually the second quantization map*

$$a_{j_1}^\dagger \dots a_{j_N}^\dagger a_{k_P} \dots a_{k_M} = \Gamma(|\varphi_{j_1} \dots \varphi_{j_M}\rangle \langle \varphi_{k_1} \dots \varphi_{k_N}|), \quad (36)$$

which can be expressed as exterior multiplication by the identity operator in $\mathcal{F}$

$$\Gamma(|\varphi_{j_1} \dots \varphi_{j_M}\rangle \langle \varphi_{k_1} \dots \varphi_{k_N}|) = |\varphi_{j_1} \dots \varphi_{j_M}\rangle \langle \varphi_{k_1} \dots \varphi_{k_N}| \wedge I_{\mathcal{H}_{\mathcal{F}}}. \quad (37)$$

The expansion and contraction maps transform naturally between the first and second quantization direct sum decompositions of $\mathcal{F}$ i.e.

$$\Gamma : \bigoplus_{0 \leq N, M \leq r} \mathcal{B}(\mathcal{H}^N, \mathcal{H}^M) \rightarrow \bigoplus_{0 \leq N, M \leq r} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{N, M} \quad (38)$$

and

$$L : \bigoplus_{0 \leq N, M \leq r} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{N, M} \rightarrow \bigoplus_{0 \leq N, M \leq r} \mathcal{B}(\mathcal{H}^N, \mathcal{H}^M). \quad (39)$$

C. Reduced Density Operators

The P^{th} order reduced density operator associated with a state $D \in S_{\mathcal{F}}$ can be defined by using the contraction map L as

$$D_P = P_{\mathcal{H}^P} L(D) P_{\mathcal{H}^P} = \sum_{\substack{1 \leq j_1 < \dots < j_P \leq r \\ 1 \leq k_1 < \dots < k_P \leq r}} \left(a_{k_1}^\dagger \dots a_{k_P}^\dagger a_{j_1} \dots a_{j_P} \middle| D \right) |\varphi_{j_1} \dots \varphi_{j_P}\rangle \langle \varphi_{k_1} \dots \varphi_{k_P}| \quad (40)$$

and in particular the SORDO is given by

$$D_2 = P_{\mathcal{H}^2} L(D) P_{\mathcal{H}^2} = \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} \left(a_{k_1}^\dagger a_{k_2}^\dagger a_{j_1} a_{j_2} \middle| D \right) |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}|. \quad (41)$$

If D is a state describing a system of N electrons then

$$\begin{aligned} D &= D^N = P_{\mathcal{H}^N} D P_{\mathcal{H}^N} \\ P_{\mathcal{H}^M} D P_{\mathcal{H}^M} &= 0, M \neq N \end{aligned} \quad (42)$$

and the above construction corresponds to the normalization

$$\text{Tr} \{D_2\} = \binom{N}{2}. \quad (43)$$

IV. EXTENSIONS OF POSITIVE FUNCTIONALS

The Fermi Fock algebra is a concrete realization of a w^* -algebra [9] (that contains an identity element), which is a c^* -algebra that also has the property of being a dual space of another Banach space, in this case $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ is the dual space of $\mathcal{B}_1(\mathcal{H}_{\mathcal{F}})$. The important property of w^* -algebras vis-à-vis c^* -algebras is that they are the closure of their idempotents, which in the case of $\mathcal{F}$ are the orthogonal projectors onto subspaces of $\mathcal{H}_{\mathcal{F}}$.

A. States

A state on a w^* -algebra is a bounded positive linear function, f , on the algebra such that $f(I) = 1$, where I is the identity element of the algebra, it is called a normal state if it can be identified with an element of the predual of the algebra otherwise it is an abnormal state. In this paper only normal states belonging to the predual $\mathcal{B}_1(\mathcal{H}_{\mathcal{F}})$ of $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ are considered.

In the Fermi Fock algebra normal states are in one to one correspondence with normalized density operators i.e.

$$f(X) = \text{Tr} \{X D_f\}; D_f \in S_{\mathcal{F}}. \quad (44)$$

B. Partial States

A *partial* state is a positive linear functional defined on a self adjoint subspace, $\mathcal{W}$, of a w^* -algebra *that contains* the identity I and is a restriction of a state. It can be shown [10, 16] that the necessary and sufficient condition for a positive linear functional defined on such a subspace $\mathcal{W}$ to be a partial state and thus extended to a state is

$$\sup_{\substack{I \geq X \geq 0 \\ X \in \mathcal{W}}} f(X) \leq 1 \quad (45)$$

and

$$\inf_{\substack{X \geq I \\ X \in \mathcal{W}}} f(X) \geq 1. \quad (46)$$

The identity element $I_{\mathcal{H}_{\mathcal{F}}}$ of $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ can be expanded as

$$I_{\mathcal{H}_{\mathcal{F}}} = \sum_{0 \leq N \leq r} P_{\mathcal{H}^N}, \quad (47)$$

where $P_{\mathcal{H}^N}$ are the orthogonal projectors onto $\mathcal{H}^N$, $0 \leq N \leq r$ and in particular $P_{\mathcal{H}^0} = |\phi\rangle\langle\phi|$ is the projector onto the vacuum state. Thus the condition eq.(46) only effects operators that are non zero on the vacuum state.

C. Reduced States

A *reduced* state is a positive linear functional defined on a self adjoint subspace of a w^* -algebra *that does not contain* the identity and is a restriction of a state. The necessary and sufficient condition [11] for a linear functional defined on such a subspace $\mathcal{W}$ to be a reduced state and thus be extended to a state is

$$1 \geq f(X) \geq 0, \quad \forall X \in \mathfrak{B}_{1+}(\mathcal{W}) \quad (48)$$

where

$$\mathfrak{B}_{1+}(\mathcal{W}) = \{X | X \in \mathcal{W}, I \geq X \geq 0\} \quad (49)$$

and

$$\inf_{X \in \mathfrak{A}_{1+}(\mathcal{W})} f(X) \geq 1, \quad (50)$$

where

$$\mathfrak{A}_{1+}(\mathcal{W}) = \{X \mid X \in \mathcal{W}, X \geq I\}. \quad (51)$$

An extension of an important theorem of Kadison [13] applied to self adjoint subspaces of w^* -algebra shows that the positive extension condition eq.(49) can be expressed in terms of projectors belonging $\mathcal{W}$, an observation that is crucial in deriving necessary and sufficient conditions for representability (Section V).

In order to obtain representability conditions for SORDO's the appropriate subspace to consider in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ is the subspace formed by one and two particle conserving operators

$$\Lambda_2 = \bigoplus_{1 \leq P \leq 2} \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_P}, \quad (52)$$

as reduced states correspond to SORDO's through

$$D_2 = \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1}\right) |\varphi_{j_1} \varphi_{j_2}\rangle \langle \varphi_{k_1} \varphi_{k_2}|, \quad (53)$$

where

$$f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1}\right) = \left(a_{k_1}^\dagger a_{k_2}^\dagger a_{j_2} a_{j_1} \middle| D\right) \quad (54)$$

and FORDO's through

$$D_1 = \sum_{\substack{1 \leq j_1 \leq r \\ 1 \leq k_1 \leq r}} f\left(a_{j_1}^\dagger a_{k_1}\right) |\varphi_{j_1}\rangle \langle \varphi_{k_1}|, \quad (55)$$

where

$$f\left(a_{j_1}^\dagger a_{k_1}\right) = \left(a_{k_1}^\dagger a_{j_1} \middle| D\right). \quad (56)$$

In $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ the condition eq.(51) only effects operators that are non zero on the vacuum state thus does not constrain any operators belonging to Λ_2 and one can just concentrate on the conditions given by eq.(49).

V. PROJECTORS

As we are searching for necessary and sufficient conditions for extending positive *linear* functionals the condition eq. (5) needs only to be checked on extreme points of the positive

portion of the unit ball $\mathfrak{B}_1(\Lambda_2)$, which are projection operators [13]. One should note that these conditions will be representability conditions and that further constraints need to be considered (section VIII) to obtain N -representability conditions. Projectors in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$ are positive idempotent operators, P , with norm equal to one i.e.

$$P \geq 0 \quad (57a)$$

$$P^2 = P \quad (57b)$$

$$\|P\|_{\mathcal{B}(\mathcal{H}_{\mathcal{F}})} = \sup_{|\Psi\rangle \in \mathcal{H}_{\mathcal{F}}} \left\{ \frac{\langle \Psi | P \Psi \rangle}{\langle \Psi | \Psi \rangle} \right\} = 1. \quad (57c)$$

We first characterize the projectors in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$, then $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2}$ and finally Λ_2 .

A. Projectors in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$

Every positive operator in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$ can be expressed as (Appendix B)

$$\lambda_1 = \sum_{1 \leq j, k \leq r} \lambda_{1jk} a_j^\dagger a_k = \sum_{1 \leq l \leq r} \beta_l b_l^\dagger b_l, \quad \beta_l \geq 0, 1 \leq l \leq r, \quad (58)$$

where

$$b_l^\dagger = \sum_{1 \leq j \leq r} u_{lj} a_j^\dagger, \quad 1 \leq l \leq r \quad (59)$$

and the coefficients $\{u_{lj}\}$ form a $r \times r$ unitary matrix i.e.

$$\mathbb{U}^\dagger \mathbb{U} = \mathbb{U} \mathbb{U}^\dagger = \mathbb{I}_r. \quad (60)$$

In Appendix B we show that the idempotency condition

$$\left(\sum_{1 \leq l \leq r} \beta_l b_l^\dagger b_l \right)^2 = \sum_{1 \leq l \leq r} \beta_l b_l^\dagger b_l \quad (61)$$

together with the norm condition

$$\left\| \sum_{1 \leq l \leq r} \beta_l b_l^\dagger b_l \right\|_{\mathcal{B}(\mathcal{H}_{\mathcal{F}})} = 1 \quad (62)$$

can only be satisfied by positive operators where

$$\beta_l = 0, \quad 1 \leq l \neq k \leq r \quad (63)$$

for some fixed k . Thus all projectors belonging to $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$ are of the form

$$\lambda_1 = b^\dagger b \quad (64)$$

where $b^\dagger$ creates a normalized General Spin Orbital (GSO)

B. Projectors in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2}$

Every positive operator in $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2}$ can be expressed as (Appendix B)

$$\lambda_2 = \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} \lambda_{2jklm} a_j^\dagger a_k^\dagger a_m a_l = \sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l, \quad \beta_l \geq 0, 1 \leq l \leq \binom{r}{2}, \quad (65)$$

where the geminal creator is

$$g_l^\dagger = \sum_{1 \leq j < k \leq r} u_{ljk} a_j^\dagger a_k^\dagger, \quad 1 \leq l \leq \binom{r}{2} \quad (66)$$

and the coefficients $\{u_{2ljk} | 1 \leq l \leq \binom{r}{2}, 1 \leq j < k \leq r\}$ form a $\binom{r}{2} \times \binom{r}{2}$ unitary matrix i.e.

$$\mathbb{U}^\dagger \mathbb{U} = \mathbb{U} \mathbb{U}^\dagger = \mathbb{I}_{\binom{r}{2}}. \quad (67)$$

In Appendix C we show that the idempotency condition

$$\left(\sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l \right)^2 = \sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l \quad (68)$$

together with the norm condition

$$\left\| \sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l \right\|_{\mathcal{B}(\mathcal{H}_{\mathcal{F}})} = 1 \quad (69)$$

can only be satisfied by positive operators where

$$\beta_l = 0, \quad 1 \leq l \neq k \leq r \quad (70)$$

for some fixed k and the rank of the k^{th} geminal must be one. Thus in order that the two particle operator λ_2 be a projection operator it must be of the form

$$\lambda_2 = a^\dagger b^\dagger b a \quad (71)$$

where $a^\dagger$ and $b^\dagger$ create normalized orthogonal GSO's.

C. Projectors in Λ_2

1. Projections from Two Hole Operators

By considering the positivity, idempotency and normalization relationship for two *hole* operators

$$\eta_2 = \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} \eta_{2jklm} a_j a_k a_m^\dagger a_l^\dagger = \sum_{1 \leq l \leq \binom{r}{2}} \gamma_l h_l h_l^\dagger, \quad \gamma_l \geq 0, 1 \leq l \leq \binom{r}{2}, \quad (72)$$

where the geminal annihilator is

$$g_l = \sum_{1 \leq j < k \leq r} u_{2ljk} a_j a_k, \quad 1 \leq l \leq \binom{r}{2} \quad (73)$$

and the coefficients $\{u_{2ljk} | 1 \leq l \leq \binom{r}{2}, 1 \leq j < k \leq r\}$ form a $\binom{r}{2} \times \binom{r}{2}$ unitary matrix i.e.

$$\mathbf{u}_2^\dagger \mathbf{u}_2 = \mathbf{u}_2 \mathbf{u}_2^\dagger = \mathbb{I}_{\binom{r}{2}}, \quad (74)$$

one can carry out the same analysis as in subsection VB and obtain (Appendix C) that in order that the two hole operator η_2 be a projection operator it must be of the form

$$\eta_2 = b a a^\dagger b^\dagger, \quad (75)$$

where a and b annihilate normalized orthogonal GSO's. The projection operator η_2 does not belong to $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2}$ but by using the canonical anticommutation relationships eq. (20) one observe that

$$\eta_2 = b a a^\dagger b^\dagger = I_{\mathcal{H}_{\mathcal{F}}} - a^\dagger a - b^\dagger b + a^\dagger b^\dagger b a \quad (76)$$

and thus

$$a^\dagger a + b^\dagger b - a^\dagger b^\dagger b a = I_{\mathcal{H}_{\mathcal{F}}} - \eta_2, \quad (77)$$

which shows that

$$\tilde{\lambda}_2 = a^\dagger a + b^\dagger b - a^\dagger b^\dagger b a \quad (78)$$

is also a projection operator belonging to Λ_2 .

2. Projections from Products of One Particle Operators

By considering the positivity, idempotency and normalization relationships for products $\lambda_1 \lambda_1'$ of one particle operators of the form eq. (58) one can determine (Appendix C) that

these products are projection operators when

$$\lambda_1 \lambda'_1 = a^\dagger b b^\dagger a = a^\dagger a - a^\dagger b^\dagger b a \in \Lambda_2. \quad (79)$$

One can prove (Appendix C) that there are no other projection operators of the form

$$\lambda = \sum_{1 \leq j, k \leq r} \lambda_{1jk} a_j^\dagger a_k + \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} \lambda_{2jklm} a_j^\dagger a_k^\dagger a_m a_l \quad (80)$$

other than

$$a^\dagger a, a^\dagger b^\dagger b a, a^\dagger a - a^\dagger b^\dagger b a, a^\dagger a + b^\dagger b - a^\dagger b^\dagger b a. \quad (81)$$

VI. GENERALIZED P, Q AND G CONDITIONS

The Necessary conditions P, Q and G conditions for representability of a $\binom{r}{2} \times \binom{r}{2}$ hermitian matrices were quickly discovered [14] and are bases independent i.e. invariant under the unitary group $U(\mathcal{H}^1)$ of one electron space. In the notation of this paper they are:

A. P-Condition

A potential Second Order Reduced Density Matrix (SORDM) matrix $\mathbb{D}_2$ given by

$$D_{2j_1 j_2 k_1 k_2} = \left(a_{k_1}^\dagger a_{k_2}^\dagger a_{j_1} a_{j_2} \middle| D \right) = \overline{D_{2k_1 k_2 j_1 j_2}} \quad (82)$$

is semi definite i.e.

$$\mathbb{D}_2 \geq 0. \quad (83)$$

The basic variables of the ground state energy optimization problem are the complex matrix elements $\{ D_{2j_1 j_2 k_1 k_2} | 1 \leq j_1 < j_2 \leq r, 1 \leq k_1 < k_2 \leq r \}$.

B. Q-Condition

The matrix $\mathbb{Q}$ given by

$$Q_{j_1 j_2 k_1 k_2} = \left(a_{k_1}^\dagger a_{j_1} a_{k_2}^\dagger a_{j_2} \middle| D \right) = \overline{Q_{k_1 k_2 j_1 j_2}} \quad (84)$$

is semi definite i.e.

$$\mathbb{Q} \geq 0. \quad (85)$$

If one invokes the pure N condition (Section VIII) this condition becomes a necessary N -representability condition and the $\mathbb{Q}$ matrix can be expressed in terms of the matrix $\mathbb{D}_2$ as

$$Q_{j_1 j_2 k_1 k_2} = \frac{1}{2(N-1)} \sum_{1 \leq l \leq r} D_{2j_2 l k_1 l} \delta_{j_1 k_2} - D_{2j_1 j_2 k_1 k_2}. \quad (86)$$

C. G-Condition

The matrix $\mathbb{G}_2$ given by

$$G_{j_1 j_2 k_1 k_2} = \left(a_{j_1} a_{j_2} a_{k_1}^\dagger a_{k_2}^\dagger \middle| D \right) = \overline{G_{2k_1 k_2 j_1 j_2}} \quad (87)$$

is semi definite i.e.

$$\mathbb{G} \geq 0. \quad (88)$$

Again if one invokes the pure N condition (Section VIII) this condition becomes a necessary N -representability condition and the $\mathbb{G}$ matrix can be expressed in terms of the matrix $\mathbb{D}_2$ as

$$G_{j_1 j_2 k_1 k_2} = \frac{1}{2(N-1)} \sum_{1 \leq l \leq r} \{ D_{2j_2 l k_1 l} \delta_{j_1 k_2} + D_{2j_1 l k_2 l} \delta_{j_2 k_1} \} - D_{2j_1 j_2 k_1 k_2}. \quad (89)$$

The generalized P, Q and G conditions are *Necessary and Sufficient* conditions for the existence of positive extensions of bounded linear functionals

$$f : \Lambda_2 \rightarrow \mathbb{C} \quad (90)$$

to a states of $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$, that are based on the theorems of Krien [10], Segal [11] and Kadison [13].

D. Generalized P-Condition

A linear functional f can be extended to a state if

$$1 \geq f(a^\dagger b^\dagger b a) \geq 0 \forall a, b, \quad (91)$$

where $|a\rangle$ and $|b\rangle$ are normalized orthogonal GSO's belonging to $\mathcal{H}^1$ and $a^\dagger$ and $b^\dagger$ are the creators of those states. As

$$D_{2abab} = f(a^\dagger b^\dagger b a) \quad (92)$$

this is a *necessary* condition for ensemble representability.

E. Generalized Q-Condition

A linear functional f can be extended to a state if

$$1 \geq f(a^\dagger a - a^\dagger b^\dagger b a) \geq 0 \forall a, b, \quad (93)$$

which gives

$$1 \geq D_{1aa} - D_{2abab} \geq 0 \forall a, b, \quad (94)$$

where the diagonal elements of a possible First Order Reduced Density Matrix (FORDM) are defined as

$$D_{1aa} = f(a^\dagger a). \quad (95)$$

Again this is a *necessary* condition for ensemble representability.

F. Generalized G-Condition

A linear functional f can be extended to a state if

$$1 \geq f(a^\dagger a + b^\dagger b - a^\dagger b^\dagger b a) \geq 0 \forall a, b, \quad (96)$$

which gives

$$1 \geq D_{1aa} - D_{1bb} - D_{2abab} \geq 0 \forall a, b \quad (97)$$

and also a *necessary* condition for ensemble representability.

VII. PARTICLE AND PAIR NUMBER CONDITIONS

In order to obtain N -representable extension one must constrain the functional f to be dispersion free on the number operator i.e.

$$f(\mathcal{N}_1^2) = f(\mathcal{N}_1)^2, \quad (98)$$

where

$$\mathcal{N}_1 = \sum_{1 \leq j \leq r} a_j^\dagger a_j. \quad (99)$$

One can show (Appendix E) that eq.(98) is equivalent to

$$f(\mathcal{N}_1) = N \quad (100)$$

and

$$f(\mathcal{N}_2) = \binom{N}{2} \quad (101)$$

for some fixed N , where the pair operator $\mathcal{N}_2$ is defined as

$$\mathcal{N}_2 = \sum_{1 \leq j < k \leq r} a_j^\dagger a_k^\dagger a_k a_j. \quad (102)$$

If f is dispersion free on the number operator the values of f on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$ must be related to its values on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2}$ by

$$f(a_j^\dagger a_k) = (N-1) f(a_j^\dagger \mathcal{N}_1 a_k) = (N-1) \sum_{1 \leq l \leq r} f(a_j^\dagger a_l^\dagger a_l a_k), \quad (103)$$

thus if the functional is only defined on $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e2}$ one can define allowable extensions to $\mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e1}$ by eq. (103).

In terms of matrix elements the definition eq. (103) becomes

$$D_{1jk} = (N-1) \sum_{1 \leq l \leq r} D_{2jlk} \quad (104)$$

and one can see that if the value of the functional on the pair operator is set by eq. (101) then the particle number condition eq. (100) is automatically ensured.

VIII. N-REPRESENTABILITY CONDITIONS

The representability conditions in section VI can be complimented and modified by the conditions in section VII to form ensemble *Necessary and Sufficient* N -representability conditions which are:

1. Normalization

$$\sum_{1 \leq j < k \leq r} D_{2jkjk} = \binom{N}{2} \quad (105)$$

2. Generalized P-Condition

$$1 \geq D_{2abab} \geq 0 \quad \forall a, b \quad (106)$$

3. Generalized Q-Condition

$$1 \geq (N-1) \sum_{1 \leq l \leq r} D_{2alal} - D_{2abab} \geq 0 \quad \forall a, b \quad (107)$$

4. Generalized G-Condition

$$1 \geq (N-1) \sum_{1 \leq l \leq r} \{D_{2alal} + D_{2blbl}\} - D_{2abab} \geq 0 \forall a, b. \quad (108)$$

IX. DISCUSSION

APPENDIX A: APPENDIX A

APPENDIX B: APPENDIX B

In the following the Fock space $\mathcal{H}_{\mathcal{F}}$ will be considered as the underlying Hilbert space on which the operators are defined and a few basic properties will be listed.

1. Finite Rank Operators

An operator X is a finite rank operator acting $\mathcal{H}_{\mathcal{F}}$ if

$$X = \sum_{1 \leq j \leq m < \infty} \alpha_j |\Phi_j\rangle \langle \Psi_j|, \quad (B1)$$

where $\{|\Phi_j| 1 \leq j \leq m\}$ and $\{|\Psi_j| 1 \leq j \leq m\}$ are orthonormal sets in $\mathcal{H}_{\mathcal{F}}$. The set of finite rank operators is denoted $\mathcal{B}_0(\mathcal{H}_{\mathcal{F}})$ and is not closed.

2. Bounded Operators

The operator norm (also called the uniform norm) is defined as

$$\|X\| = \sup_{|\Phi\rangle \in \mathcal{H}_F} \left\{ \left(\frac{\langle \Phi | X^\dagger X \Phi \rangle}{\langle \Phi | \Phi \rangle} \right)^{\frac{1}{2}} \right\} = \sup_{|\Phi\rangle \in \mathcal{H}_F} \left\{ \frac{\|X\Phi\|_{\mathcal{H}_F}}{\|\Phi\|_{\mathcal{H}_F}} \right\}, \quad (B2)$$

(where $\|\cdot\|_{\mathcal{H}_F}$ is the norm in $\mathcal{H}_F$), and the space of bounded operators acting on the Hilbert space $\mathcal{H}_{\mathcal{F}}$ as

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}}) = \{X \mid \|X\| < \infty\}. \quad (B3)$$

If one restricts attention to self adjoint operators $X = X^\dagger$ then it can be shown that

$$\|X\| = \sup_{|\Phi\rangle \in \mathcal{H}_F} \left\{ \left| \frac{\langle \Phi | X \Phi \rangle}{\langle \Phi | \Phi \rangle} \right| \right\} \quad (B4)$$

and thus for positive operators, $X \geq 0$

$$\|X\| = \sup_{|\Phi\rangle \in \mathcal{H}_F} \left\{ \frac{\langle \Phi | X \Phi \rangle}{\langle \Phi | \Phi \rangle} \right\}. \quad (\text{B5})$$

The space $\mathcal{B}(\mathcal{H}_F)$ is closed in the operator norm topology.

3. Compact Operators

The space of compact operators, $\mathcal{C}(\mathcal{H}_F)$, is the closed space produced by taking the closure of the set of finite rank operators with respect to the operator norm topology i.e. any compact operator can be expressed as

$$X = \lim_{m \rightarrow \infty} \sum_{1 \leq j \leq m} \alpha_j |\Phi_j\rangle \langle \Psi_j|, \quad (\text{B6})$$

in the sense that

$$\lim_{m \rightarrow \infty} \left\| X - \sum_{1 \leq j \leq m} \alpha_j |\Phi_j\rangle \langle \Psi_j| \right\| = 0. \quad (\text{B7})$$

4. Trace Class Operators

The trace class norm is defined as

$$\|X\|_1 = \text{Tr} \left\{ (X^\dagger X)^{\frac{1}{2}} \right\} \quad (\text{B8})$$

and the space of trace class operators acting on the Hilbert space $\mathcal{H}_F$ as

$$\mathcal{B}_1(\mathcal{H}_F) = \{X \mid \|X\|_1 < \infty\}. \quad (\text{B9})$$

The space $\mathcal{B}_1(\mathcal{H}_F)$ is closed with respect to the $\|\cdot\|_1$ norm, while the closure with respect to the $\|\cdot\|$ norm produces $\mathcal{C}(\mathcal{H}_F)$.

5. Hilbert-Schmidt Operators

The Hilbert-Schmidt norm is defined as

$$\|X\|_2 = \left(\text{Tr} \{ X^\dagger X \} \right)^{\frac{1}{2}} \quad (\text{B10})$$

and the space of trace class operators acting on the Hilbert space $\mathcal{H}_{\mathcal{F}}$ as

$$\mathcal{B}_2(\mathcal{H}_{\mathcal{F}}) = \{X \mid \|X\|_2 < \infty\}. \quad (\text{B11})$$

The space $\mathcal{B}_2(\mathcal{H}_{\mathcal{F}})$ is closed with respect to the $\|\cdot\|_2$ norm, while the closure with respect to $\|\cdot\|$ norm produces $\mathcal{C}(\mathcal{H}_{\mathcal{F}})$.

6. Relationship between the Spaces

The relationship between the spaces is

$$\mathcal{B}(\mathcal{H}_{\mathcal{F}}) \supseteq \mathcal{C}(\mathcal{H}_{\mathcal{F}}) \supseteq \mathcal{B}_2(\mathcal{H}_{\mathcal{F}}) \supseteq \mathcal{B}_1(\mathcal{H}_{\mathcal{F}}) \supseteq \mathcal{B}_0(\mathcal{H}_{\mathcal{F}}). \quad (\text{B12})$$

The closure of $\mathcal{B}_0(\mathcal{H}_{\mathcal{F}})$ (and thus of $\mathcal{C}(\mathcal{H}_{\mathcal{F}})$, $\mathcal{B}_2(\mathcal{H}_{\mathcal{F}})$ and $\mathcal{B}_1(\mathcal{H}_{\mathcal{F}})$) with respect to the strong operator operator is $\mathcal{B}(\mathcal{H}_{\mathcal{F}})$, where these limits can be defined as

$$X = \text{s-}\lim_{m \rightarrow \infty} X_m \iff \lim_{m \rightarrow \infty} \|(X - X_m)\Phi\| = 0 \quad \forall |\Phi\rangle \in \mathcal{H}_{\mathcal{F}}, \quad (\text{B13})$$

in contrast to the uniform operator topology limits that are given by

$$X = \lim_{m \rightarrow \infty} X_m \iff \lim_{m \rightarrow \infty} \|(X - X_m)\| = 0 = \lim_{m \rightarrow \infty} \sup_{|\Phi\rangle \in \mathcal{H}_{\mathcal{F}}} \left\{ \left(\frac{\langle \Phi | (X - X_m)^\dagger (X - X_m) | \Phi \rangle}{\langle \Phi | \Phi \rangle} \right)^{\frac{1}{2}} \right\}. \quad (\text{B14})$$

If the dimension of $\mathcal{H}_{\mathcal{F}}$ is finite all of these spaces are identical and closed in all of the preceding norms and topologies.

APPENDIX C: APPENDIX C

A general element of Λ_2 can be expanded as

$$\lambda = \sum_{1 \leq j, k \leq r} \lambda_{1jk} a_j^\dagger a_k + \sum_{\substack{1 \leq j < k \leq r \\ 1 \leq l < m \leq r}} \lambda_{2jklm} a_j^\dagger a_k^\dagger a_m a_l \quad (\text{C1})$$

the condition that it is a projector i.e. idempotent is that

$$\lambda = \lambda^2. \quad (\text{C2})$$

In a c^* -algebra

$$\|X^\dagger X\| = \|X\|^2 \quad (\text{C3})$$

The idempotency condition eq. (C2) implies that

$$\|\lambda^2\| = \|\lambda\| \quad (\text{C4})$$

and the c^* norm property eq. (C3)

$$\|\lambda^2\| = \|\lambda\|^2 \quad (\text{C5})$$

gives

$$\|\lambda\|^2 = \|\lambda\| \quad (\text{C6})$$

and thus

$$\|\lambda\| = 1. \quad (\text{C7})$$

The idempotency condition eq. (C2) also implies that

$$\lambda \geq 0. \quad (\text{C8})$$

1. One particle case

We first consider the case $\lambda_2 = 0$ then

$$\lambda_1 = \lambda_1^2 \quad (\text{C9})$$

and noting that λ_1 must be self adjoint and positive one has

$$\lambda = \lambda_1 = \sum_{1 \leq j, k \leq \infty} \lambda_{1jk} a_j^\dagger a_k = \sum_{\substack{1 \leq j, k \leq r \\ 1 \leq l \leq r}} u_{jl} \beta_l \bar{u}_{kl} a_j^\dagger a_k = \sum_{1 \leq l \leq r} \beta_l b_l^\dagger b_l, \quad (\text{C10})$$

where

$$\beta_l \geq 0; 1 \leq j \leq r \quad (\text{C11a})$$

$$b_l^\dagger = \sum_{1 \leq j \leq r} a_j^\dagger u_{jl} \quad (\text{C11b})$$

$$b_l^\dagger |\phi\rangle = |\psi_l\rangle \quad (\text{C11c})$$

$$\mathbb{U}^\dagger \mathbb{U} = \mathbb{U} \mathbb{U}^\dagger = \mathbb{I}_r. \quad (\text{C11d})$$

The condition eq. (C9) becomes

$$\sum_{1 \leq l \leq r} \beta_l b_l^\dagger b_l = \sum_{1 \leq l \leq r} \beta_l^2 b_l^\dagger b_l + 2 \sum_{1 \leq l_1 < l_2 \leq r} \beta_{l_1} \beta_{l_2} b_{l_1}^\dagger b_{l_1} b_{l_2}^\dagger b_{l_2}, \quad (\text{C12})$$

which can be rewritten as

$$\sum_{1 \leq l \leq r} \{\beta_l (1 - \beta_l)\} b_l^\dagger b_l = 2 \sum_{1 \leq l_1 < l_2 \leq r} \beta_{l_1} \beta_{l_2} b_{l_1}^\dagger b_{l_2}^\dagger b_{l_2} b_{l_1}. \quad (\text{C13})$$

This equality must be true on all vectors in $\mathcal{H}_{\mathcal{F}}$ in particular on $|\psi_\kappa\rangle \in \mathcal{H}^1, 1 \leq \kappa \leq r$, i.e.

$$\sum_{1 \leq l \leq r} \{\beta_l (1 - \beta_l)\} b_l^\dagger b_l |\psi_1\rangle = 2 \sum_{1 \leq l_1 < l_2 \leq r} \beta_{l_1} \beta_{l_2} b_{l_1}^\dagger b_{l_2}^\dagger b_{l_2} b_{l_1} |\psi_1\rangle, \quad (\text{C14})$$

which gives

$$\beta_\kappa (1 - \beta_\kappa) |\psi_\kappa\rangle = 0, \quad 1 \leq \kappa \leq r \quad (\text{C15})$$

i.e.

$$\beta_\kappa = 0 \text{ or } \beta_\kappa = 1, \quad 1 \leq \kappa \leq r. \quad (\text{C16})$$

One can see that $\beta_\kappa = 1$ for only one value of $\kappa \in \{1, \dots, r\}$ for let $\{n_1, \dots, n_{r'}\}$ be a subset of $\{1, \dots, r\}$ and

$$\lambda_1 = \sum_{\sigma \in \{1, \dots, r\}} b_\sigma^\dagger b_\sigma \quad (\text{C17})$$

then

$$\lambda_1^2 = \sum_{\sigma \in \{1, \dots, r\}} b_\sigma^\dagger b_\sigma + \sum_{\sigma, \sigma' \in \{1, \dots, r\}} b_\sigma^\dagger b_{\sigma'}^\dagger b_{\sigma'} b_\sigma = \lambda_1 = \sum_{\sigma \in \{1, \dots, r\}} b_\sigma^\dagger b_\sigma \quad (\text{C18})$$

thus

$$\sum_{\sigma, \sigma' \in \{1, \dots, r\}} b_\sigma^\dagger b_{\sigma'}^\dagger b_{\sigma'} b_\sigma = 0, \quad (\text{C19})$$

which is only possible if $r' = 1$. Thus if λ_1 it must be of the form

$$\lambda_1 = b^\dagger b. \quad (\text{C20})$$

2. Two particle case

We consider the case $\lambda_1 = 0$ then

$$\lambda_2 = \lambda_2^2 \quad (\text{C21})$$

and

$$\lambda = \lambda_2 = \sum_{1 \leq j, k \leq \infty} \lambda_{2j_1 j_2 k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} = \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r \\ 1 \leq l \leq \binom{r}{2}}} u_{j_1 j_2 l} \beta_l \bar{u}_{l k_1 k_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_2} a_{k_1} = \sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l, \quad (\text{C22})$$

where

$$\beta_l \geq 0; 1 \leq l \leq \binom{r}{2}, \quad (\text{C23a})$$

$$g_l^\dagger = \sum_{1 \leq j_1 < j_2 \leq r} u_{lj_1 j_2} a_{j_1}^\dagger a_{j_2}^\dagger = \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} u_{lj_1 j_2} a_{j_1}^\dagger a_{j_2}^\dagger; u_{lj_1 j_2} = -u_{lj_2 j_1} \quad (\text{C23b})$$

$$g_l^\dagger |\phi\rangle = |g_l\rangle, \langle g_{l_1} | g_{l_2} \rangle = \delta_{l_1 l_2} \quad (\text{C23c})$$

$$\mathbb{U}_2^\dagger \mathbb{U}_2 = \mathbb{U}_2 \mathbb{U}_2^\dagger = \mathbb{I}_{\binom{r}{2}} \equiv \text{Tr} \left\{ \mathbb{U}_{1l_1}^\dagger \mathbb{U}_{1l_2} \right\} = \delta_{l_1 l_2}, 1 \leq l_1, l_2 \leq r \quad (\text{C23d})$$

$$\mathbb{U}_{1l_2} = -\mathbb{U}_{1l_2}^t \quad (\text{C23e})$$

The condition eq. (C21) becomes

$$\sum_{1 \leq l \leq \binom{r}{2}} \beta_l g_l^\dagger g_l = \sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} g_{l_1}^\dagger g_{l_1} g_{l_2}^\dagger g_{l_2}, \quad (\text{C24})$$

$$= \sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} g_{l_1}^\dagger \left([g_{l_1}, g_{l_2}^\dagger] + g_{l_2}^\dagger g_{l_1} \right) g_{l_2} \quad (\text{C25})$$

$$= \sum_{1 \leq l_1, l_2 \leq \binom{r}{2}} \beta_{l_1} \beta_{l_2} \left\{ g_{l_1}^\dagger [g_{l_1}, g_{l_2}^\dagger] g_{l_2} + g_{l_1}^\dagger g_{l_2}^\dagger g_{l_1} g_{l_2} \right\} \quad (\text{C26})$$

$$[g_{l_1}, g_{l_2}^\dagger] = - \sum_{\substack{1 \leq j_1 < j_2 \leq r \\ 1 \leq k_1 < k_2 \leq r}} \left\{ \bar{u}_{l_1 j_1 j_2} u_{l_2 j_2 j_1} - \bar{u}_{l_1 j_1 j_2} u_{l_2 j_1 j_2} + \bar{u}_{l_1 j_1 j_2} u_{l_2 j_1 k_2} a_{k_2}^\dagger a_{j_2} - \bar{u}_{l_1 j_1 j_2} u_{l_2 k_1 j_1} a_{k_1}^\dagger a_{j_2} \right. \quad (\text{C27})$$

$$\left. - \bar{u}_{l_1 j_1 j_2} u_{l_2 j_2 k_2} a_{k_2}^\dagger a_{j_1} + \bar{u}_{l_1 j_1 j_2} u_{l_2 k_1 j_2} a_{k_1}^\dagger a_{j_1} \right\} \quad (\text{C28})$$

$$[g_{l_1}, g_{l_2}^\dagger] = \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \bar{u}_{l_1 j_1 j_2 l_1} u_{l_2 k_1 k_2} [a_{j_2} a_{j_1}, a_{k_1}^\dagger a_{k_2}^\dagger] \quad (\text{C29})$$

$$= -\frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \bar{u}_{l_1 j_1 j_2} u_{l_2 k_1 k_2} [a_{k_1}^\dagger a_{k_2}^\dagger, a_{j_2} a_{j_1}] \quad (\text{C30})$$

$$= -\frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \bar{u}_{l_1 j_1 j_2} u_{l_2 k_1 k_2} \{ I_{\mathcal{H}_{\mathcal{F}}} \delta_{j_2 k_1} \delta_{j_1 k_2} - I_{\mathcal{H}_{\mathcal{F}}} \delta_{j_2 k_2} \delta_{j_1 k_1} \} \quad (\text{C31})$$

$$a_{k_2}^\dagger a_{j_2} \delta_{j_1 k_1} - a_{k_1}^\dagger a_{j_2} \delta_{j_1 k_2} - a_{k_2}^\dagger a_{j_1} \delta_{j_2 k_1} + a_{k_1}^\dagger a_{j_1} \delta_{j_2 k_2} \} \quad (\text{C32})$$

$$= -\frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \left\{ -\bar{u}_{l_1 j_1 j_2} u_{l_2 j_1 j_2} - \bar{u}_{l_1 j_1 j_2} u_{l_2 j_1 j_2} + \bar{u}_{l_1 j_2 j_1} u_{l_2 k_2 j_1} a_{k_2}^\dagger a_{j_2} + \bar{u}_{l_1 j_2 j_1} u_{l_2 k_1 j_1} a_{k_1}^\dagger a_{j_2} \right. \quad (\text{C33})$$

$$\left. + \bar{u}_{l_1 j_1 j_2} u_{l_2 k_2 j_2} a_{k_2}^\dagger a_{j_1} + \bar{u}_{l_1 j_1 j_2} u_{l_2 k_1 j_2} a_{k_1}^\dagger a_{j_1} \right\} \quad (\text{C34})$$

$$= -\frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} \left\{ -\delta_{l_1 l_2} - \delta_{l_1 l_2} + \left(\mathbb{U}_{1l_1}^\dagger \mathbb{U}_{1l_2} \right)_{j_2 k_2} a_{k_2}^\dagger a_{j_2} + \left(\mathbb{U}_{1l_1}^\dagger \mathbb{U}_{1l_2} \right)_{j_2 k_1} a_{k_1}^\dagger a_{j_2} \right. \quad (\text{C35})$$

$$\left. + \left(\mathbb{U}_{1l_1}^\dagger \mathbb{U}_{1l_2} \right)_{j_1 k_2} a_{k_2}^\dagger a_{j_1} + \left(\mathbb{U}_{1l_1}^\dagger \mathbb{U}_{1l_2} \right)_{j_1 k_1} a_{k_1}^\dagger a_{j_1} \right\} \quad (\text{C36})$$

$$= \frac{1}{2} \delta_{l_1 l_2} - \sum_{1 \leq j_1, k_1 \leq r} \left(\mathbb{U}_{1l_1}^\dagger \mathbb{U}_{1l_2} \right)_{j_1 k_1} a_{k_1}^\dagger a_{j_1} \quad (\text{C37})$$

First one must determine the commutator

$$\left[a_{j_1}^\dagger a_{j_2}^\dagger, a_{j_4} a_{j_3} \right] \quad (\text{C38})$$

thus one brings $a_{j_4} a_{j_3} a_{j_1}^\dagger a_{j_2}^\dagger$ to normal form as

$$\begin{aligned} a_{j_4} a_{j_3} a_{j_1}^\dagger a_{j_2}^\dagger &= a_{j_4} \left(I \delta_{j_3 j_1} - a_{j_1}^\dagger a_{j_3} \right) a_{j_2}^\dagger = a_{j_4} a_{j_2}^\dagger \delta_{j_3 j_1} - a_{j_4} a_{j_1}^\dagger a_{j_3} a_{j_2}^\dagger = \\ &\left(\delta_{j_4 j_2} - a_{j_2}^\dagger a_{j_4} \right) \delta_{j_3 j_1} - \left(I \delta_{j_4 j_1} - a_{j_1}^\dagger a_{j_4} \right) \left(I \delta_{j_3 j_2} - a_{j_2}^\dagger a_{j_3} \right) = \\ &I \delta_{j_4 j_2} \delta_{j_3 j_1} - a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} - I \delta_{j_4 j_1} \delta_{j_3 j_2} + a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} + a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} - a_{j_1}^\dagger a_{j_4} a_{j_2}^\dagger a_{j_3} = \\ &I \delta_{j_4 j_2} \delta_{j_3 j_1} - a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} - I \delta_{j_4 j_1} \delta_{j_3 j_2} + a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} + a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} - a_{j_1}^\dagger \left(\delta_{j_4 j_2} - a_{j_2}^\dagger a_{j_4} \right) a_{j_3} = \\ &I \delta_{j_4 j_2} \delta_{j_3 j_1} - I \delta_{j_4 j_1} \delta_{j_3 j_2} - a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} + a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} + a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} - a_{j_1}^\dagger a_{j_3} \delta_{j_4 j_2} + a_{j_1}^\dagger a_{j_2}^\dagger a_{j_4} a_{j_3}, \end{aligned} \quad (\text{C39})$$

hence

$$\begin{aligned} a_{j_4} a_{j_3} a_{j_1}^\dagger a_{j_2}^\dagger &= \\ &I \delta_{j_4 j_2} \delta_{j_3 j_1} - I \delta_{j_4 j_1} \delta_{j_3 j_2} - a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} + a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} + a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} - a_{j_1}^\dagger a_{j_3} \delta_{j_4 j_2} + a_{j_1}^\dagger a_{j_2}^\dagger a_{j_4} a_{j_3} \end{aligned} \quad (\text{C40})$$

and

$$\left[a_{j_1}^\dagger a_{j_2}^\dagger, a_{j_4} a_{j_3} \right] = I \delta_{j_4 j_1} \delta_{j_3 j_2} - I \delta_{j_4 j_2} \delta_{j_3 j_1} + a_{j_2}^\dagger a_{j_4} \delta_{j_3 j_1} - a_{j_1}^\dagger a_{j_4} \delta_{j_3 j_2} - a_{j_2}^\dagger a_{j_3} \delta_{j_4 j_1} + a_{j_1}^\dagger a_{j_3} \delta_{j_4 j_2} \quad (\text{C41})$$

Theorem 1 $g^\dagger g g^\dagger g = g^\dagger g \iff g^\dagger = a_{j_k}^\dagger a_k^\dagger$, when $\|g\|_{\mathcal{H}^2}^2 = 1$ and $c_1 \neq 0$

$$g^\dagger g g^\dagger g = \sum_{1 \leq j_1, j_2, j_3, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger a_{j_2} a_{j_2+s} a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_4} a_{j_4+s} c_{j_1} c_{j_2} c_{j_3} c_{j_4} \quad (\text{C42})$$

Observing that

$$\begin{aligned}
& a_{j_2} a_{j_2+s} a_{j_3}^\dagger a_{j_3+s}^\dagger \\
&= \delta_{j_2 j_3+s} \delta_{j_2+s j_3} - \delta_{j_2 j_3} \delta_{j_2+s j_3+s} - a_{j_3+s}^\dagger a_{j_2} \delta_{j_2+s j_3} + \\
& \quad + a_{j_3}^\dagger a_{j_2} \delta_{j_2+s j_3+s} + a_{j_3+s}^\dagger a_{j_2+s} \delta_{j_2 j_3} - a_{j_3}^\dagger a_{j_2+s} \delta_{j_2 j_3+s} + a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2} \\
&= -I \delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_2} \delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_2} \delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2}
\end{aligned} \tag{C43}$$

one obtains

$$\begin{aligned}
g^\dagger g g^\dagger g &= \sum_{1 \leq j_1, j_2, j_3, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger \left(-\delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_2} \delta_{j_2 j_3} + a_{j_3+s}^\dagger a_{j_2+s} \delta_{j_2 j_3} + a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2} \right) \\
& \quad \times a_{j_4} a_{j_4+s} c_{j_1} c_{j_2} c_{j_3} c_{j_4} \\
&= \sum_{1 \leq j_1, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger a_{j_4+s} a_{j_4} c_{j_1} c_{j_4} \|g\|_{\mathcal{H}^2}^2 \\
& \quad + \sum_{1 \leq j_1, j_2, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger \left(a_{j_2}^\dagger a_{j_2} + a_{j_2+s}^\dagger a_{j_2+s} \right) a_{j_4} a_{j_4+s} c_{j_1} n_{j_2} c_{j_4} \\
& \quad + \sum_{1 \leq j_1, j_2, j_3, j_4 \leq s} a_{j_1}^\dagger a_{j_1+s}^\dagger a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2} a_{j_4} a_{j_4+s} c_{j_1} c_{j_2} c_{j_3} c_{j_4}.
\end{aligned} \tag{C44}$$

Hence

$$\begin{aligned}
& g^\dagger g g^\dagger g - g^\dagger g = 0 \Rightarrow \\
& a_{j_1}^\dagger a_{j_1+s}^\dagger \left(a_{j_2}^\dagger a_{j_2} + a_{j_2+s}^\dagger a_{j_2+s} \right) a_{j_4} a_{j_4+s} = 0 \\
& a_{j_1}^\dagger a_{j_1+s}^\dagger a_{j_3}^\dagger a_{j_3+s}^\dagger a_{j_2+s} a_{j_2} a_{j_4} a_{j_4+s} = 0
\end{aligned} \tag{C45}$$

and the only solution is $s = 1$. Consider $s = 2$ then e.g. and the only solution is $s = 1$.

Consider $s = 2$ then e.g.

$$\begin{aligned}
& a_1^\dagger a_3^\dagger \left(a_2^\dagger a_2 + a_4^\dagger a_4 \right) a_1 a_3 n_1 n_2 = 0 \\
& a_1^\dagger a_3^\dagger a_2^\dagger a_4^\dagger a_3 a_1 a_2 a_4 n_1 n_2 = 0,
\end{aligned} \tag{C46}$$

The generalized Q -condition is based on the product

$$A^\dagger A, \tag{C47}$$

where

$$A = a_{j_1}^\dagger a_{j_2}^\dagger, \tag{C48}$$

i.e.

$$A^\dagger A = a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger. \quad (\text{C49})$$

This operator is a projector as

$$\begin{aligned} \left(a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \right)^2 &= a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger = a_{j_2} a_{j_1} a_{j_1}^\dagger \left(1 - a_{j_2} a_{j_2}^\dagger \right) a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger = \\ &= a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger - a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2} a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \\ &= a_{j_2} a_{j_1} \left(1 - a_{j_1} a_{j_1}^\dagger \right) a_{j_1}^\dagger a_{j_2}^\dagger + a_{j_2} a_{j_1} a_{j_2} a_{j_1}^\dagger a_{j_2}^\dagger a_{j_1}^\dagger a_{j_2}^\dagger \\ &= a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger - a_{j_2} a_{j_1} a_{j_1} a_{j_1}^\dagger a_{j_1}^\dagger a_{j_2}^\dagger \\ &= a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger. \end{aligned} \quad (\text{C50})$$

Noting that

$$a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger = I_{\mathcal{H}_{\mathcal{F}}} - a_{j_2}^\dagger a_{j_2} - a_{j_1}^\dagger a_{j_1} + a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \quad (\text{C51})$$

one has

$$f \left(a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \right) = 1 - f \left(a_{j_2}^\dagger a_{j_2} \right) - f \left(a_{j_1}^\dagger a_{j_1} \right) + f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \quad (\text{C52})$$

and if f is a state then

$$\begin{aligned} 1 &\geq f \left(a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \right) \geq 0 \Leftrightarrow 1 \geq 1 - f \left(a_{j_2}^\dagger a_{j_2} \right) - f \left(a_{j_1}^\dagger a_{j_1} \right) + f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \geq 0 \\ &\Leftrightarrow 0 \geq f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) - f \left(a_{j_2}^\dagger a_{j_2} \right) - f \left(a_{j_1}^\dagger a_{j_1} \right) \geq -1 \\ &\Leftrightarrow 1 \geq f \left(a_{j_2}^\dagger a_{j_2} \right) + f \left(a_{j_1}^\dagger a_{j_1} \right) - f \left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1} \right) \geq 0 \end{aligned} \quad (\text{C53})$$

One should note however that

$$a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \notin \Lambda_2 = \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_1} \oplus \mathcal{B}(\mathcal{H}_{\mathcal{F}})_{e_2}, \quad (\text{C54})$$

but

$$I_{\mathcal{H}_{\mathcal{F}}} - a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \quad (\text{C55})$$

does, as

$$I_{\mathcal{H}_{\mathcal{F}}} - a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger = a_{j_1}^\dagger a_{j_1} + a_{j_2}^\dagger a_{j_2}, \quad (\text{C56})$$

which is a projector as

$$\begin{aligned} \left(I_{\mathcal{H}_{\mathcal{F}}} - a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \right)^2 &= I_{\mathcal{H}_{\mathcal{F}}} - 2a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger + \left(a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger \right)^2 \\ &= I_{\mathcal{H}_{\mathcal{F}}} - a_{j_2} a_{j_1} a_{j_1}^\dagger a_{j_2}^\dagger. \end{aligned} \quad (\text{C57})$$

This condition can be formulated as

$$1 \geq f\left(\Phi(\chi)^\dagger \Phi(\chi)\right) + f\left(\Phi(\psi)^\dagger \Phi(\psi)\right) - f\left(\Phi(\chi)^\dagger \Phi(\psi)^\dagger \Phi(\chi) \Phi(\psi)\right) \geq 0$$

$$\forall \langle \chi | \psi \rangle = 0; \langle \psi | \psi \rangle = \langle \chi | \chi \rangle = 1 \quad (\text{C58})$$

The generalized G-condition is based on the product

$$A^\dagger A, \quad (\text{C59})$$

where

$$A = a_{j_1}^\dagger a_{j_2}, \quad (\text{C60})$$

i.e.

$$A^\dagger A = a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2} = a_{j_2}^\dagger \left(I_{\mathcal{H}_{\mathcal{F}}} - a_{j_1}^\dagger a_{j_1} \right) a_{j_2} = a_{j_2}^\dagger a_{j_2} - a_{j_2}^\dagger a_{j_1}^\dagger a_{j_1} a_{j_2}. \quad (\text{C61})$$

The operator $a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2} = n_{j_1} n_{j_2}$ is a projector as

$$n_{j_1} n_{j_2} n_{j_1} n_{j_2} = n_{j_1} n_{j_1} n_{j_2} n_{j_2} = n_{j_1} n_{j_2} \quad (\text{C62})$$

and

$$a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2} \in \Lambda_2. \quad (\text{C63})$$

If f is a state then the Generalized G-condition gives

$$1 \geq f\left(a_{j_2}^\dagger a_{j_1} a_{j_1}^\dagger a_{j_2}\right) = f\left(a_{j_2}^\dagger \left(I - a_{j_1}^\dagger a_{j_1}\right) a_{j_2}\right) = f\left(a_{j_2}^\dagger a_{j_2} - a_{j_2}^\dagger a_{j_1}^\dagger a_{j_1} a_{j_2}\right) \geq 0$$

$$\Leftrightarrow 1 \geq f\left(a_{j_2}^\dagger a_{j_2}\right) - f\left(a_{j_1}^\dagger a_{j_2}^\dagger a_{j_2} a_{j_1}\right) \geq 0. \quad (\text{C64})$$

APPENDIX D: APPENDIX D

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$$\left[a_{k_1}^\dagger a_{k_2}^\dagger, a_{j_2} a_{j_1} \right] = I \delta_{j_2 k_1} \delta_{j_1 k_2} - I \delta_{j_2 k_2} \delta_{j_1 k_1} + a_{k_2}^\dagger a_{j_2} \delta_{j_1 k_1} - a_{k_1}^\dagger a_{j_2} \delta_{j_1 k_2} - a_{k_2}^\dagger a_{j_1} \delta_{j_2 k_1} + a_{k_1}^\dagger a_{j_1} \delta_{j_2 k_2} \quad (\text{E1})$$